

Generators of Spin $1/2$ Rotations $S_j = \frac{\hbar}{2} \sigma_j$

Unitary Rotation transformation in Hilbert Space

$$D(\hat{n}, \phi) = \exp\left(-i \frac{\vec{S} \cdot \hat{n} \phi}{\hbar}\right) = \exp\left(-i \frac{\vec{\sigma} \cdot \hat{n} \phi}{2}\right)$$

Write the Z-basis representation of

1.) $|\psi\rangle = e^{-iX\frac{\alpha}{2}} |+\rangle_y$

Show it agrees with Expectations for $\alpha = \pi/2$

Possible Approach

- Insert $\mathbb{I}_x$
- Apply operator in HS Eigenbasis
- Apply $\mathbb{I}_z$

Last Class, showed $U_{x \rightarrow y} = S$ i.e., $S|+\rangle_x = |+\rangle_y$
and

$$S|-\rangle_x = |-\rangle_y$$

$$S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/2} \end{pmatrix}$$

Use the continuous Rotation $D(\hat{z}, \alpha)$ to Express S :

$$e^{-iZ\frac{\alpha}{2}}$$

[if unsure, Act on $|+\rangle_x$

Aside:

$$T = \sqrt{S} = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$$

Describe why T is ^{historically} called " $\pi/8$ " gate, how does it modify Azimuthal Angle

$$T|\psi(\theta, \phi)\rangle = |\psi(\theta, \phi)\rangle$$

3.) $U_{z \rightarrow x} = U_{x \rightarrow z}$: $H = H^\dagger$ Hadamard is Hermitian.

(Eq. (8) of 11.D may be helpful)

$$\therefore H \neq D(\hat{y}, \pi/2).$$

find $\hat{n}_H$ s.t. H can be interpreted as $\phi_H = \pi$ rotation

1-) Could use $\exp(-iX\frac{\alpha}{2}) = \cos\frac{\alpha}{2} - iX\sin\frac{\alpha}{2}$ from Ex. II. D. 3

or, for Basis change practice:

notation

$$|+\rangle_z = |0\rangle \quad |-\rangle_z = |1\rangle$$

$$|\pm\rangle_y = |\pm i\rangle$$

$$|\pm\rangle_x = |\pm\rangle$$

$$|7\rangle = e^{-iX\frac{\alpha}{2}} |i\rangle = e^{-iX\frac{\alpha}{2}} \mathbb{1}_x |i\rangle$$

expand

$$\mathbb{1}_x = \hat{P}_{x,+} + \hat{P}_{x,-} = |+\rangle\langle+| + |-\rangle\langle-|$$

$$|7\rangle = e^{-iX\frac{\alpha}{2}} \left(|+\rangle\langle+|i\rangle + |-\rangle\langle-|i\rangle \right) \quad \text{complex \#s}$$

operator is diagonal in its Eigenbasis, $\Rightarrow$ eigenstate-dependent phases

$$|7\rangle = e^{-i\frac{\alpha}{2}} |+\rangle\langle+|i\rangle + e^{i\frac{\alpha}{2}} |-\rangle\langle-|i\rangle$$

Apply $\mathbb{1}_z$ to Expand in Z-basis

$$\hat{P}_{z,+} + \hat{P}_{z,-} = |0\rangle\langle 0| + |1\rangle\langle 1|$$

$$|7\rangle = |0\rangle \left(e^{-i\frac{\alpha}{2}} \underbrace{\frac{1}{\sqrt{2}}}_{\langle 0|+\rangle} \underbrace{\frac{1+i}{2} = \frac{1}{\sqrt{2}} e^{i\pi/4}}_{\langle +|i\rangle} + e^{i\frac{\alpha}{2}} \underbrace{\frac{1}{\sqrt{2}}}_{\langle 0|- \rangle} \underbrace{\frac{1-i}{2} = \frac{1}{\sqrt{2}} e^{-i\pi/4}}_{\langle -|i\rangle} \right)$$

$$+ |1\rangle \left(e^{-i\frac{\alpha}{2}} \underbrace{\frac{1}{\sqrt{2}}}_{\langle 1|+\rangle} \underbrace{\frac{1+i}{2} = \frac{1}{\sqrt{2}} e^{i\pi/4}}_{\langle +|i\rangle} + e^{i\frac{\alpha}{2}} \underbrace{\frac{1}{\sqrt{2}}}_{\langle 1|- \rangle} \underbrace{\frac{1-i}{2} = \frac{1}{\sqrt{2}} e^{-i\pi/4}}_{\langle -|i\rangle} \right)$$

Evaluating inner product complex \#s:

$$|7\rangle = \frac{|0\rangle}{2} \left(e^{-i(\frac{\alpha}{2} - \pi/4)} + e^{i(\frac{\alpha}{2} - \pi/4)} \right) + \frac{|1\rangle}{2} \left(e^{-i(\frac{\alpha}{2} - \pi/4)} - e^{i(\frac{\alpha}{2} - \pi/4)} \right)$$

for $\alpha = \pi/2$

$$= |0\rangle$$

Rotation About axis $\hat{n} = \hat{x}$ by angle $\pi/2$ takes $|i\rangle$ to $|0\rangle$ as Anticipated

$$2.) \quad S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/2} \end{pmatrix}$$

$$D(\hat{z}, \pi/2) = \exp\left(-i\frac{z\pi}{4}\right) = \begin{pmatrix} e^{-i\pi/4} & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$$

gets correct relative phase of $\pi/2$

add global phase

$$S = e^{i\pi/4} D(\hat{z}, \pi/2)$$

$$T = e^{i\pi/8} D(\hat{z}, \pi/4) = e^{i\pi/8} e^{-i\pi/8 z}$$

Historically, "forgot" $\vec{S} = \frac{1}{2} \vec{\sigma}$

$T = \pi/8$ -gate rotates by angle $\pi/4$ about z

$$3.) \quad D(\hat{n}, \phi) = \cos \phi/2 - i \vec{\sigma} \cdot \hat{n} \sin \phi/2$$

$$\text{for } \phi = \pi, \quad D(\hat{n}, \pi) = -i \vec{\sigma} \cdot \hat{n}$$

Useful for Ex. I.D.4
think of this as
Projection Coefficient

Pauli Decomposition of Hadamard:

because paulis are all traceless, square to 1, and $\text{Tr}[\mathbb{I}] = 2$, can write as:

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{\text{Tr}[H \sigma_x]}{2} \sigma_x + \frac{\text{Tr}[H \sigma_y]}{2} \sigma_y + \frac{\text{Tr}[H \sigma_z]}{2} \sigma_z$$

$$= \frac{1}{\sqrt{2}} X + \frac{1}{\sqrt{2}} Z$$

notation:

$$\sigma_x = X$$

$$\sigma_y = Y$$

$$\sigma_z = Z$$

$$\therefore H = i D(\hat{n}_H, \pi) \quad \text{with} \quad \hat{n}_H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

note: $U_{x \rightarrow z}$ not proportional to $D(\hat{y}, \pi/2)$ as perhaps inspired by #2.

Can be written as sequential product: $H = i D(\hat{x}, \pi) D(\hat{y}, \pi/2)$

"to equator by R_y then
internal phasing by 180 about
X to return to Z upon repeat

In[673]:=

```
rotationOp[nhat_,  $\phi$ _] := MatrixExp[- $\frac{i}{2}$  Sum[nhat[[jj]]  $\times$  PauliMatrix[jj], {jj, 1, 3}]  $\phi$ ];
```

1.)

In[674]:=

```
finalstate = Simplify[rotationOp[{1, 0, 0},  $\alpha$ ]. $\frac{\{1, i\}}{\sqrt{2}}$ ];

xEvecsInZ = { $\frac{\{1, 1\}}{\sqrt{2}}$ ,  $\frac{\{1, -1\}}{\sqrt{2}}$ }; (* written in Z basis*)

xprojectors = Outer[Times, #, Conjugate[#]] & /@ xEvecsInZ;
Print[" $\hat{P}_{x+} + \hat{P}_{x-} = I_x = I$  : ", Total[xprojectors] = DiagonalMatrix[{1, 1}]]
(* insertion of identityx from Z-basis. now expanded in X-basis *)

yplusinXbasis = #. $\frac{\{1, i\}}{\sqrt{2}}$  & /@ Conjugate[xEvecsInZ];

Print["| $\rightarrow$ y in X-basis: { $x\langle + | \rightarrow$ y,  $x\langle - | \rightarrow$ y} : ", yplusinXbasis]
(* operator looks like a diagonal, Z, operator in its eigenbasis (X) *)

psiinXbasis = Simplify[Exp[ $i\frac{\alpha}{2}$ ] DiagonalMatrix[{Exp[- $i\frac{\alpha}{2}$ ], Exp[ $i\frac{\alpha}{2}$ ]}].yplusinXbasis];

Print["Apply operator in X-eigenbasis: Exp[- $i\frac{\alpha}{2}$ ] ", psiinXbasis];

zbasisstates = {{1, 0}, {0, 1}};

uX2Z = Outer[Times, {1, 0},  $\frac{\{1, 1\}}{\sqrt{2}}$ ] + Outer[Times, {0, 1},  $\frac{\{1, -1\}}{\sqrt{2}}$ ];

zEvecsInX = ConjugateTranspose[uX2Z].# & /@ zbasisstates;
(* back to Z-basis vectors,  $|\psi_{inX}\rangle = |0\rangle\langle 0|\psi_{inX}\rangle + |1\rangle\langle 1|\psi_{inX}\rangle$  *)
finalpsiInZ = #.psiinXbasis & /@ Conjugate[zEvecsInX];

Print["Return to Z-basis: Exp[- $i\frac{\alpha}{2}$ ] ", finalpsiInZ]

Print["Upon  $\frac{\pi}{2}$  rotation: ", Simplify[(Exp[- $i\frac{\alpha}{2}$ ] finalpsiInZ) /. { $\alpha \rightarrow \frac{\pi}{2}$ }]]

Print["Agreement with matrix exponential? ",

finalstate == Simplify[ExpToTrig[Exp[- $i\frac{\alpha}{2}$ ] finalpsiInZ]]]
```

$$\hat{P}_{x+} + \hat{P}_{x-} = I_x = I \quad : \quad \text{True}$$

$$|+\rangle_y \text{ in X-basis: } \{ {}_x\langle + | + \rangle_y, {}_x\langle - | + \rangle_y \} : \left\{ \frac{1}{2} + \frac{i}{2}, \frac{1}{2} - \frac{i}{2} \right\}$$

$$\text{Apply operator in X-eigenbasis: } \text{Exp}\left[-i \frac{\alpha}{2}\right] \left\{ \frac{1}{2} + \frac{i}{2}, \left(\frac{1}{2} - \frac{i}{2}\right) e^{i\alpha} \right\}$$

$$\text{Return to Z-basis: } \text{Exp}\left[-i \frac{\alpha}{2}\right] \left\{ \frac{\frac{1}{2} + \frac{i}{2}}{\sqrt{2}} + \frac{\left(\frac{1}{2} - \frac{i}{2}\right) e^{i\alpha}}{\sqrt{2}}, \frac{\frac{1}{2} + \frac{i}{2}}{\sqrt{2}} - \frac{\left(\frac{1}{2} - \frac{i}{2}\right) e^{i\alpha}}{\sqrt{2}} \right\}$$

$$\text{Upon } \frac{\pi}{2} \text{ rotation: } \{1, 0\}$$

$$\text{Agreement with matrix exponential?} \quad \text{True}$$

2.)

In[689]:=

(* Sphase *)

$$\text{DiagonalMatrix}[\{1, i\}] = \text{Exp}\left[i \frac{\pi}{4}\right] \text{rotationOp}\left[\{0, 0, 1\}, \frac{\pi}{2}\right]$$

Out[689]=

True

In[690]:=

(* Tphase *)

$$\text{DiagonalMatrix}\left[\left\{1, \text{Exp}\left[i \frac{\pi}{4}\right]\right\}\right] = \text{Exp}\left[i \frac{\pi}{8}\right] \text{rotationOp}\left[\{0, 0, 1\}, \frac{\pi}{4}\right]$$

Out[690]=

True

3.)

In[691]:=

$$\text{HadamardMatrix}[2] = i \text{rotationOp}\left[\frac{\{1, 0, 1\}}{\sqrt{2}}, \pi\right]$$

Out[691]=

True

In[692]:=

$$\text{simEQtest}\left[\text{HadamardMatrix}[2], i \text{rotationOp}\left[\{1, 0, 0\}, \pi\right] \cdot \text{rotationOp}\left[\{0, 1, 0\}, \frac{\pi}{2}\right]\right]$$

Out[692]=

True